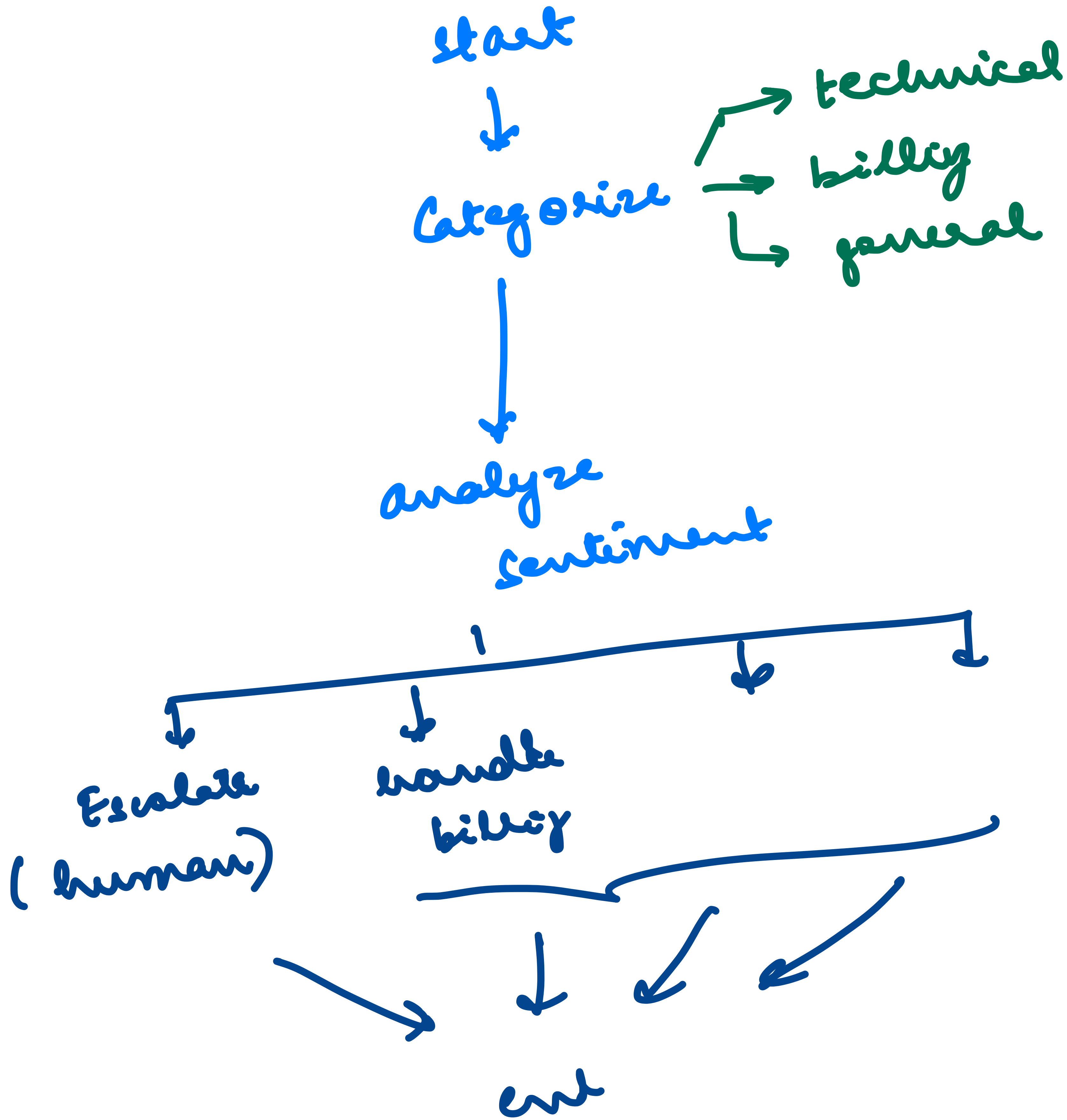

start
categorize
technical
billing
general
analyze sentiment
Escalate (human)
handle billing
end

Fine Tuning

→ Daily News

1day → Today

→ Financial   Market Analysis

Assistant

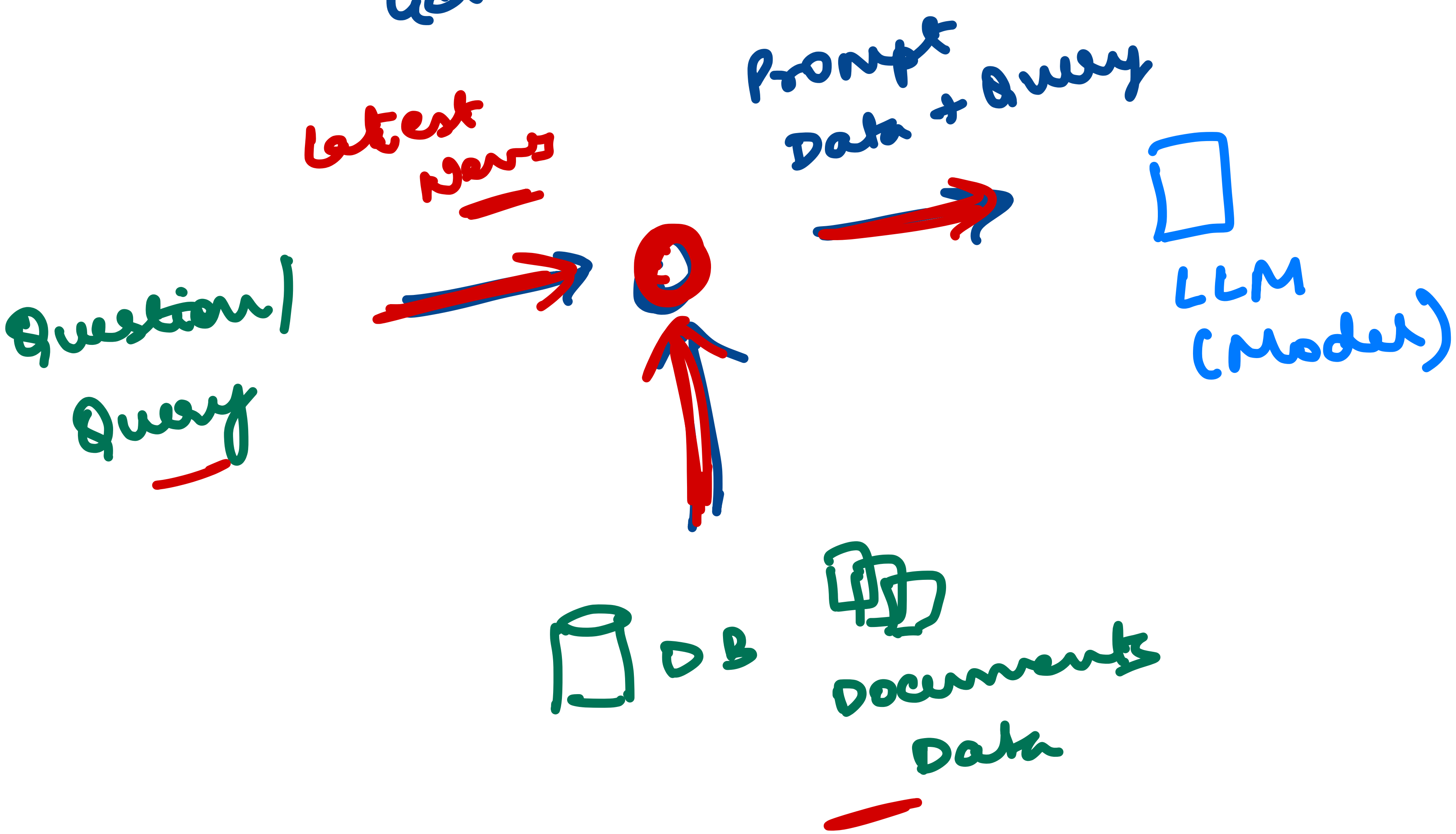
→ Tech Doc
Langgraph } → New Docs → DB

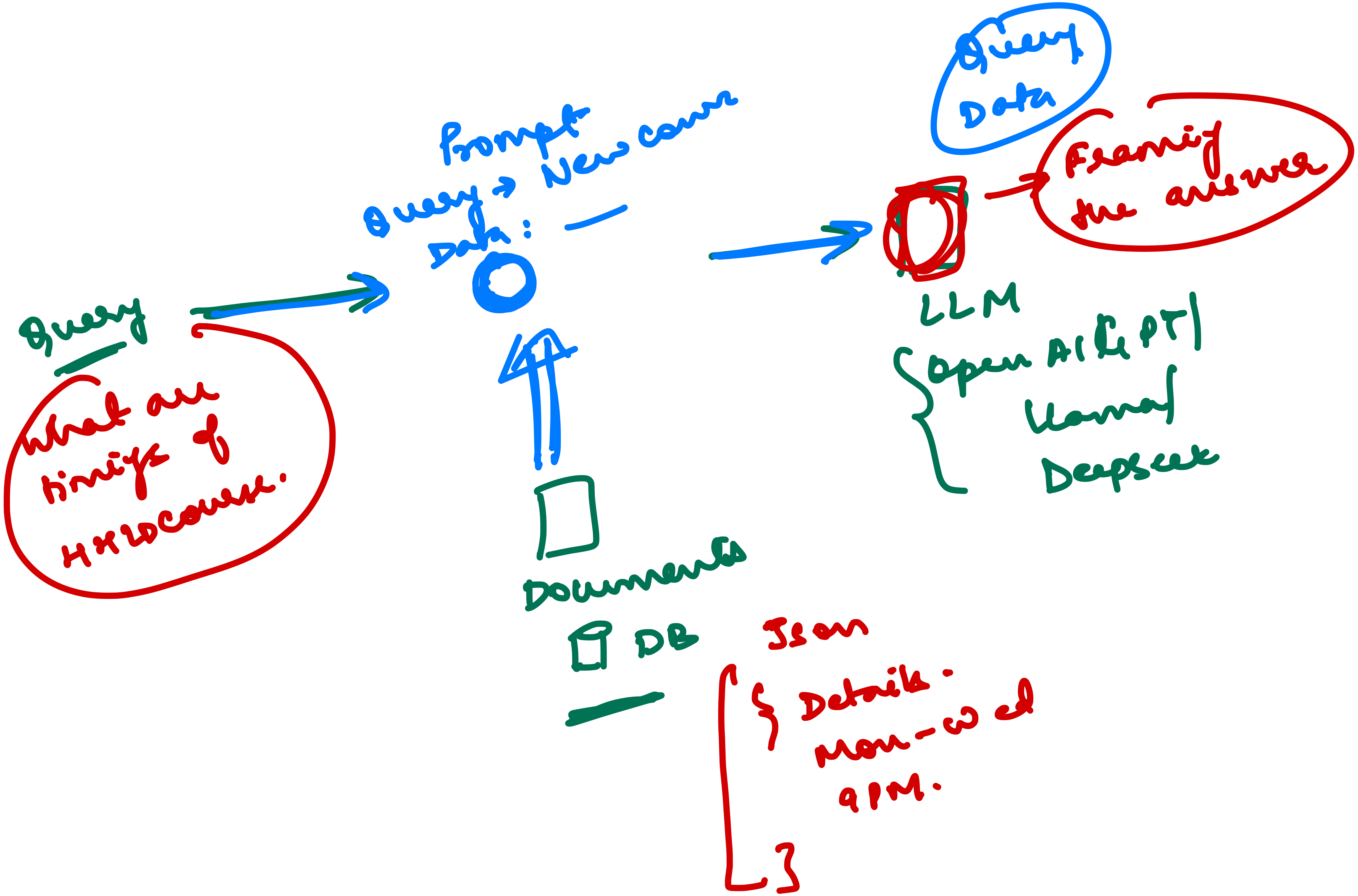
→ Products → Assistant

→ DB

RAG

Retrieval → DB
Augmented → Add
Generation → O/P




Query
What are timings of
Prompt
Query → New conv
Data :
Query
Data
Framing the answer
LLM
Open AI GPT
Llama
Deepseek
Documents
DB
Json
{ Details -
Mon - wed
9 PM.
}

Company Specific info

Employees → Manager

- Title

- Reportee

Manager

JIRA

where is my order

ID

Query + Order Details

LLM

Model

Order Status

who is the manager of Keerti → Query + Keerti's Data → LLM

Keerti DB

Fine Tuning → Parameters Training the Model

RAG → No Training

Problems

Query Time
- Fetch Data from DB
- What to fetch
- Multiple DBs → which DB
→ Amount of Data

Prompt
Query + Data
→ Tokens to Model

RAG ≠ Fine Tuning ✓
✓
New Data

Medical Diagonsis - FT

Legal Doc Drafting - FT

Products - FT

↓

RAG

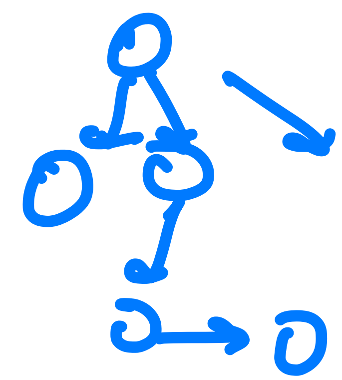
LangChain → Apps

LLM + Tools

- Agents

LangGraph → Complex

Apps / Agents



Making calls to models / LLMs

GPT | Deepseek | Llama | Claude | Qwen

Pre-trained

Train
for specific task

Fine Tuning

New Data
RAG

Query $\xrightarrow{\text{Embed}}$ $[x_1, x_2, x_3 \dots]$ → Q → LLM

$[y_1\ y_2\ y_3 \dots]$

$[z_1, z_2\ z_3 \dots]$

GenAI course

Database

Apple → [0.1 0.2 . . . -- .]

Banana → [- - - .]

course → [- - -]

AI → [- - -]

HLD Educosys → [- - -]

[ ]

Vector DB

Chroma DB

Milvus

Pinecone

Weaviate

Hash Indexes

Id →

Name →

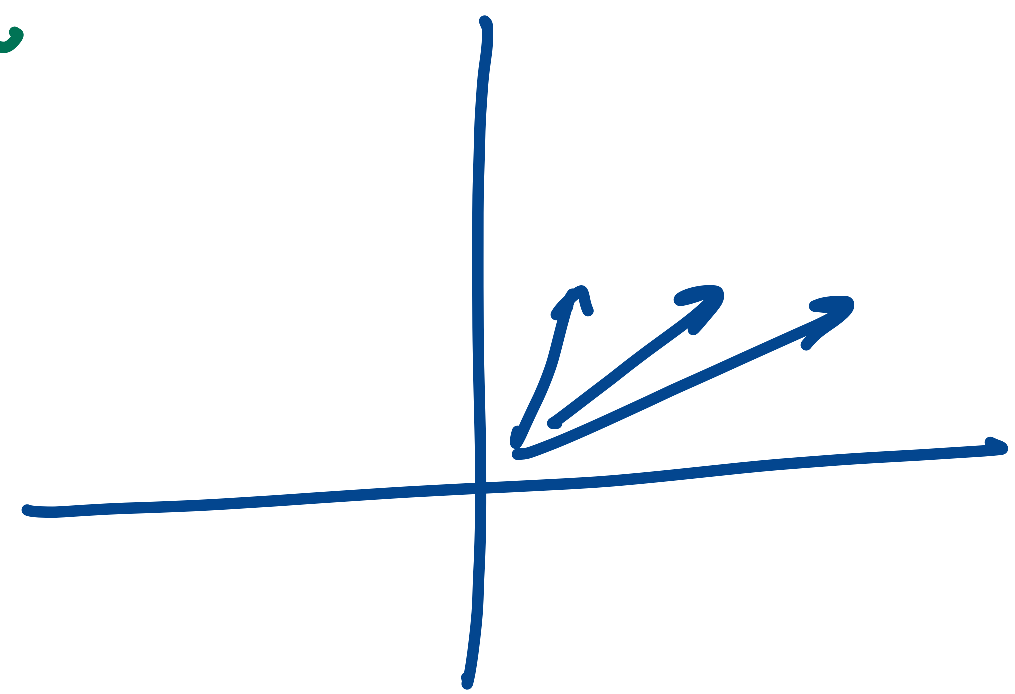
DB["Name"]

DB["ID"] → Exact Values

vectors → Search

Similarity Search

$\boxed{\text{Distances}}$



apple juice

apple juice

[0.1 0.2 0.3]

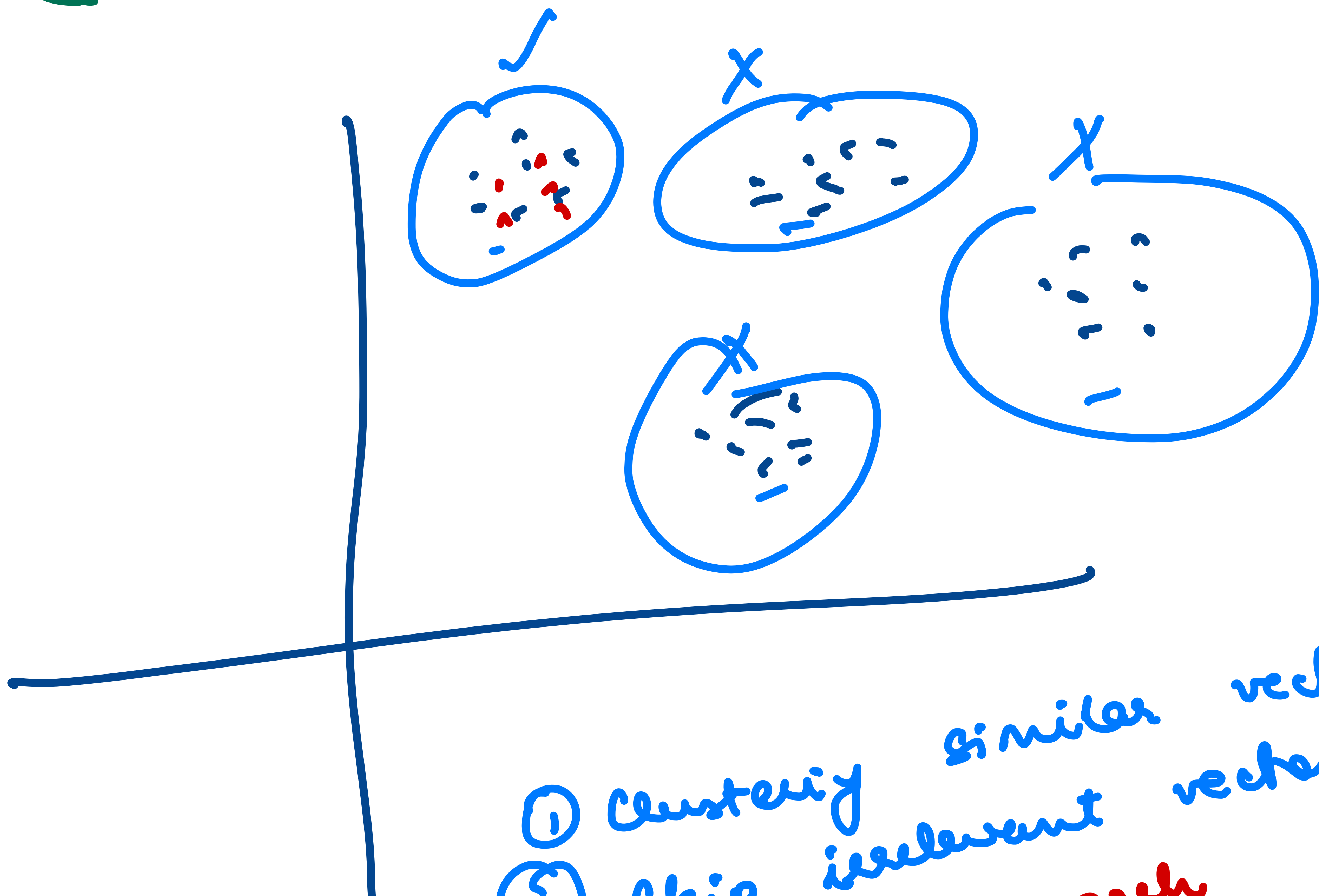
Embeddings → [Dimensions]

$$\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$$

(x, y)

$$\left[(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2 - - - \cdot \right)$$

(1000 dims



① Clustering similar vectors

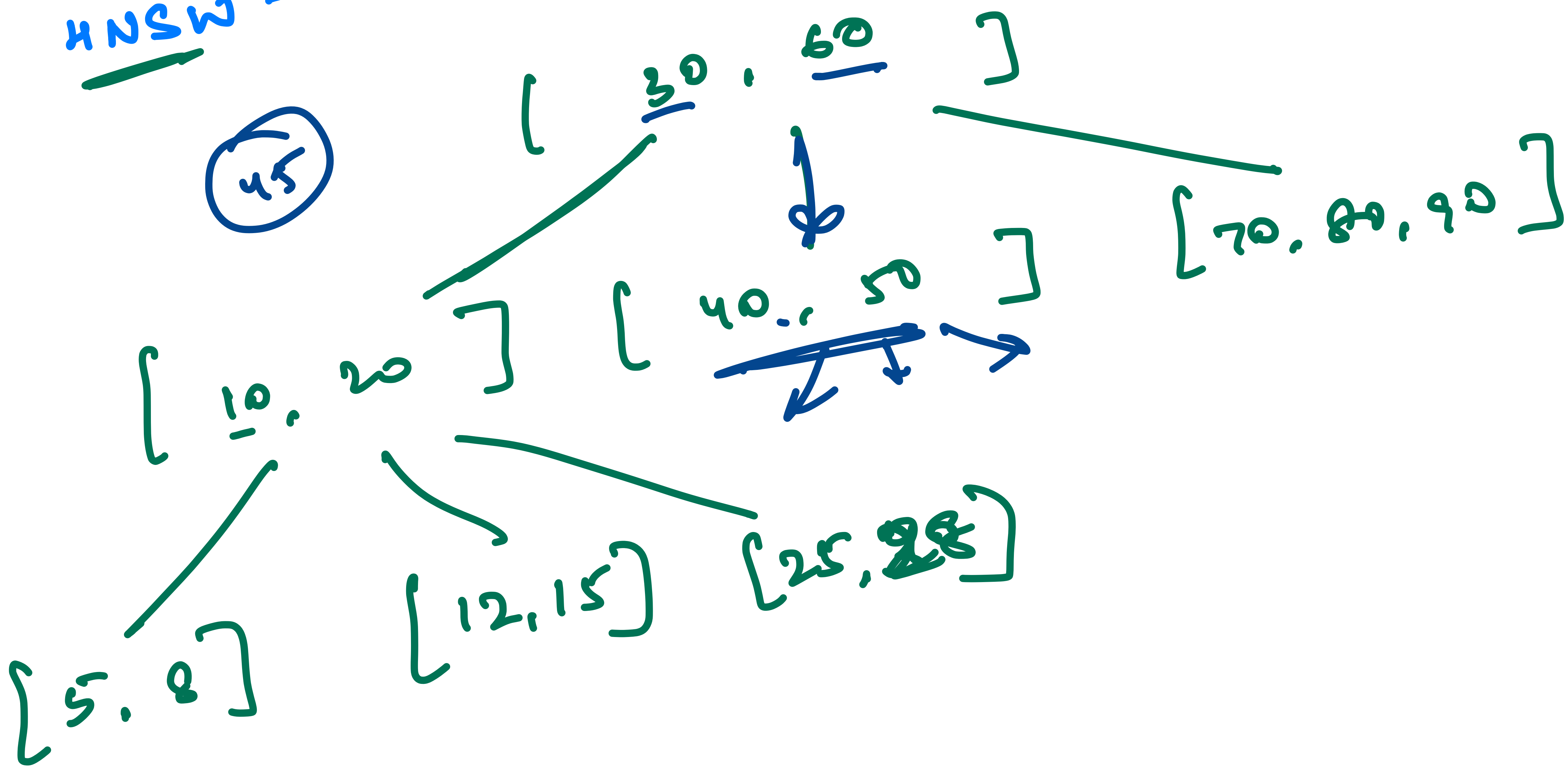
② Skip irrelevant vectors

③ Approximate search

a LOT faster but less accurate

ANN → Approximate
Nearest
Neighbors

HNSW - Hierarchial Navigable Small World



Graph [Car Dog AI Apple Red]

Banana - Mango

Bike - Bus - Truck Cat ML

Green

Grapes

Giraffe

Elephant

Boat

DL

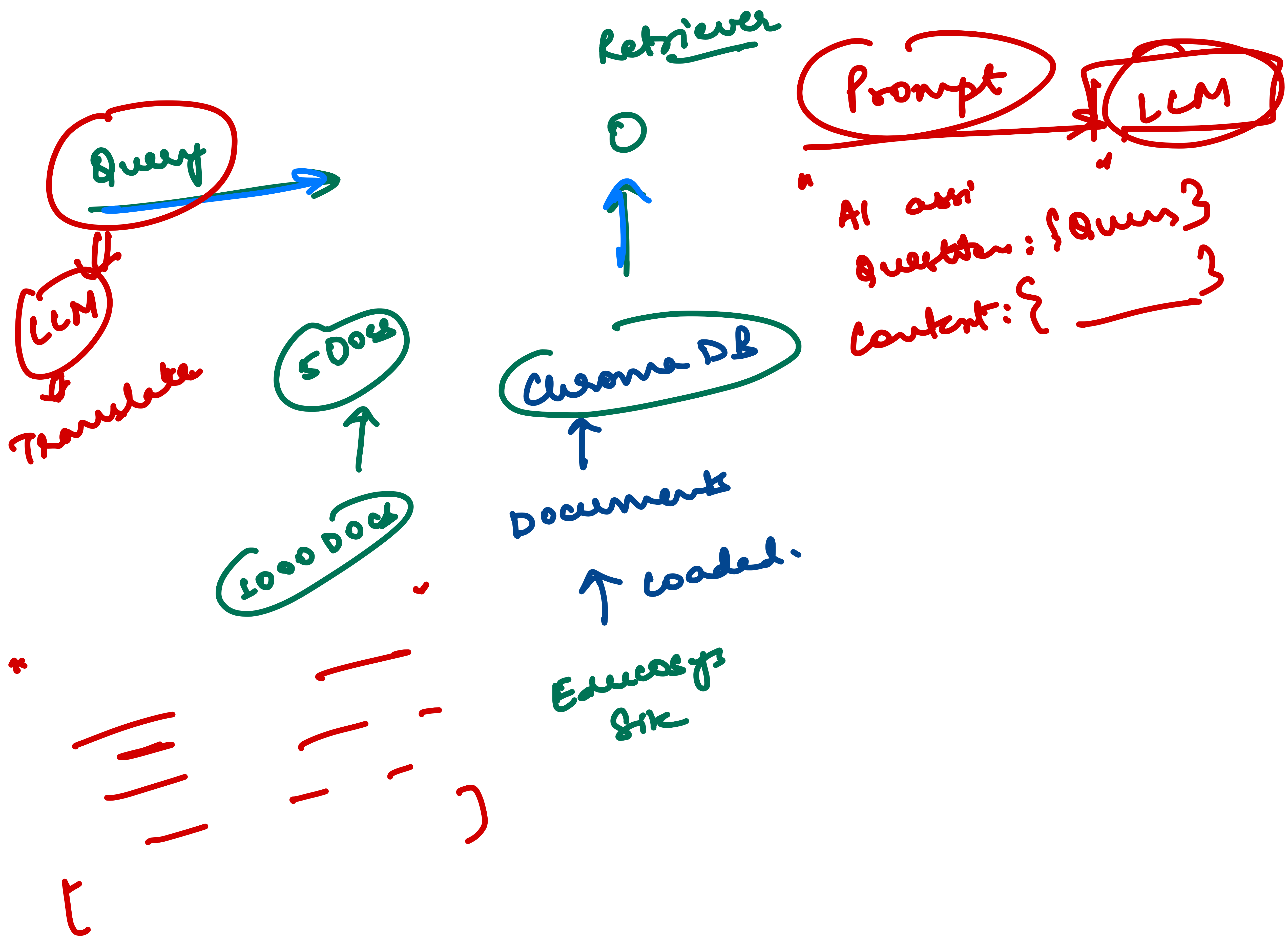
Airplane

Orange

HF

Yellow

DS Algo


Retriever
Prompt
LLM
Query
AI assi
Question: {Query}
Context: {
LLM
5 Docs
ChromaDB
Translate
Documents
1000 Docs
loaded.
Educosys
site

LLM Call.

Which DB
How to Retrieve?

LLM Call

Resp.

Query

LLM Call

+ Chinese
+ English

Query Translations

Multiple Questions

3 kinds of vector DB

SQLITE
Postgres DB

Advanced RAG
Agentic RAG

ML

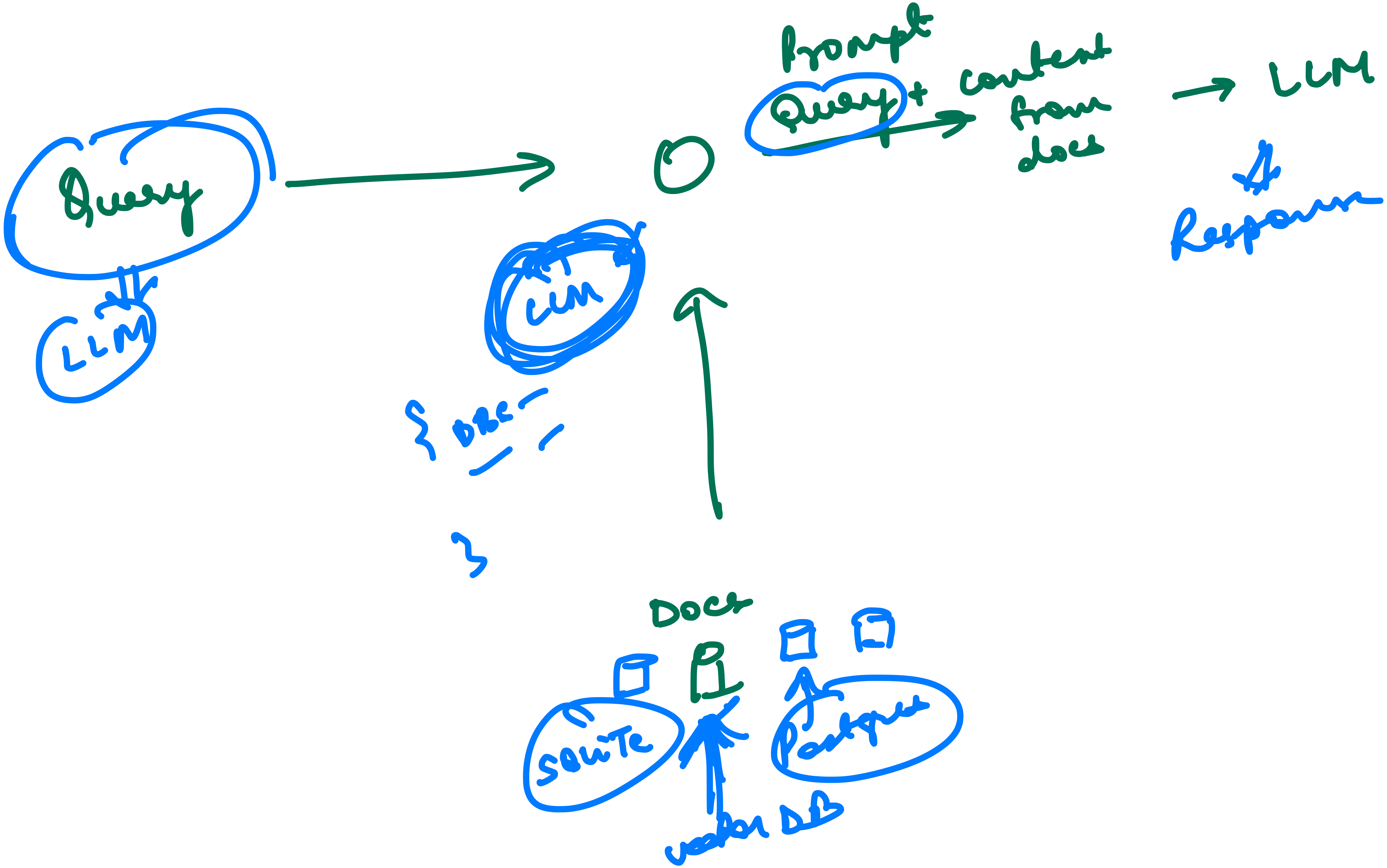
DS

Frontend

UI

Chatbot

Streamlit → Single Page


Query
LLM
LLM
{ DBs
}
Prompt
Query + content from docs
LLM
Response
Docs
SQLite
Postgres
vector DB